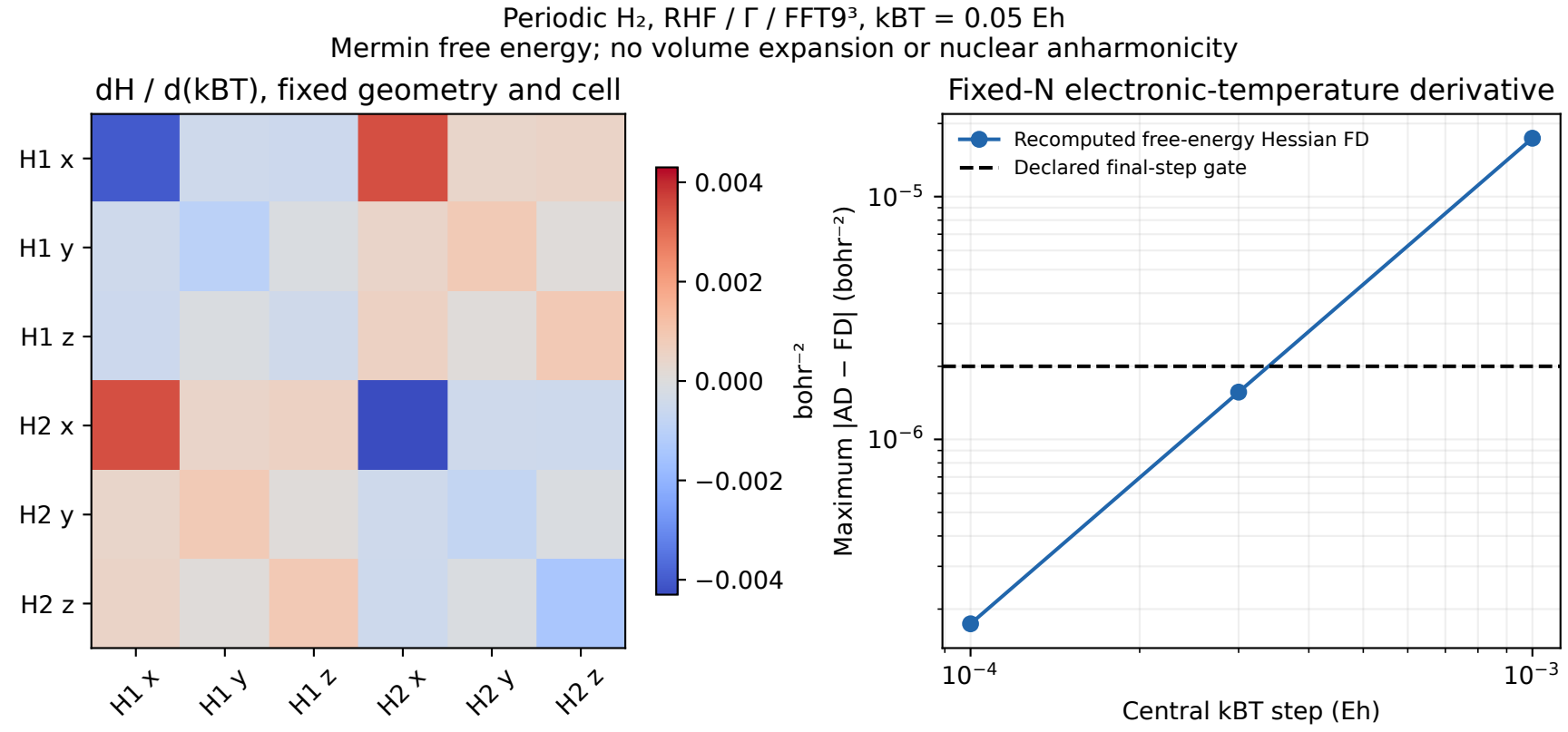
Periodic H₂, RHF / Γ / FFT9³, kBT = 0.05 Eh
Mermin free energy; no volume expansion or nuclear anharmonicity
dH / d(kBT), fixed geometry and cell
H1 x
H1 y
H1 z
H2 x
H2 y
H2 z
0.004
0.002
0.000
−0.002
−0.004
bohr⁻²
Fixed-N electronic-temperature derivative
Recomputed free-energy Hessian FD
Declared final-step gate
Maximum |AD − FD| (bohr⁻²)
10⁻⁵
10⁻⁶
10⁻⁴
10⁻³
Central kBT step (Eh)